Knowing machines with mutual truthfulness and mechanicalness

A proof of Alexander's Question 3-1-3

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Abstract

We construct a uniformly computably enumerable family of knowing machines for every computably enumerable well-founded hierarchy on the natural numbers. Each machine knows that every machine is truthful and mechanical, and knows particular program indices for the knowledge columns of its subordinates. Knowledge is interpreted by consequence in a theory of modal arithmetic, with arithmetic interpreted in the standard natural numbers. The result answers Alexander's Question 3-1-3 affirmatively for unrestricted modal formulas and hierarchies of arbitrary well-founded height. The construction uses a common auxiliary theory and a recursion-theoretic fixed point. Its soundness proof combines induction on the hierarchy with induction on finite ordinal support. Finite patterns of resemblance supply the labelled proof stages and the finite guards needed to recover source theoremhood. The same family also satisfies known mechanicalness with parameters and inherits entire subordinate theories along finite paths.

Introduction

A knowing machine is an effectively enumerable theory whose theorems are true in the model determined by the theories of all the machines. A modal operator expresses each machine's knowledge: an open formula is known at an assignment when its numerical closure follows from that machine's theory. Such a machine can reason about arithmetic, about another machine's knowledge, and about its own knowledge. The existence problem is to arrange these theories so that their claims about one another remain true despite this self-reference.

For a fixed formula with one free input variable, its **knowledge column** is the set of numbers whose numerical instances the machine knows. There are two distinct ways to describe a machine's computational nature. To know its **mechanicalness** is to know that an index exists for each unary knowledge column. To know a **particular code** is to know the column equation for a specified numerical index. The distinction is the position of the existential quantifier relative to the knowledge operator. This distinction matters when a machine reasons about itself as well as other machines.

Alexander's Question 3-1-3 [A, Definition 1.1.1] asks for machines that all know one another's truthfulness and mechanicalness, with particular codes known along a prescribed hierarchy. The hierarchy is given by a recursively enumerable well-founded relation: a lower agent is subordinate to a higher one. The relation need not be transitive, and its lower ideals need not be decidable. We prove the existence of a uniformly recursively enumerable family for every such relation. Section 1 states the three required schemas precisely; Section 2 specifies the weak modal logic and epistemic

arithmetic in which they are interpreted. We use **r.e.** as an abbreviation for recursively enumerable, equivalently computably enumerable.

The construction and its soundness proof have different roles. First, a common auxiliary theory and Kleene's recursion theorem determine all the theories and their literal program indices. This syntactic construction is completed before any theory is assumed sound. Second, temporary models interpret knowledge as raw theorem acceptance together with truth of the accepted formula. These truth filters make reflection available while soundness is still being proved. Once every raw theorem predicate is shown sound, the filters can be removed and knowledge becomes actual source consequence.

The temporary proof stages attach ordinal labels to the modal operators of finitely many agents. A cut at a label uses only generators supported below that label. A **finite guard** is a finite chain of labels related by the order-2 resemblance relation; finite-pattern reflection provides separated copies of such guards in which proofs can be placed. For each target formula, one finite room is fixed before its term variants and numerical instances are considered. Individual proof certificates can then use additional finite rooms. This quantifier order permits formulas and proof depths of arbitrary finite complexity without selecting one infinite chain of labels.

Sections 3–5 construct the source family and establish the auxiliary theory's soundness under a general outside-predicate hypothesis. Section 6 states and derives the exact ordinal input from the cited pattern-of-resemblance results. Sections 7–14 define the labelled stages, prove syntactic transport, and reconstruct source-language models. Sections 15–16 combine induction on ordinal support with well-founded induction on the hierarchy to prove soundness of the entire family. Section 17 derives the required schemas and removes the auxiliary operator. Section 18 gives further consequences and states the scope of the finite-room argument. All notation and intermediate constructions used in this proof are introduced here.

1. The theorem

Let $\mathcal{L}$ be first-order arithmetic with modal operators K_i, $i \in \mathbb{N}$, interpreted in Alexander's base logic. Fix the usual enumeration $(W_e)_{e\in\mathbb{N}}$ of r.e. subsets of $\mathbb{N}$, with a standard arithmetic formula $W(e, x)$ representing membership.

Let R be any r.e. well-founded relation on $\mathbb{N}$. The convention is that jRi means that j is subordinate to i.

Theorem. There is a uniformly r.e. family $(T_i)_{i\in\mathbb{N}}$ of closed, base-logically consequence-closed $\mathcal{L}$-theories such that the standard-arithmetic structure

$$\mathcal{M} \vDash K_i\varphi[s] \quad \Longleftrightarrow \quad T_i \vdash_{\mathrm{BL}} \varphi^s \tag{T}$$

satisfies self-doubting Epistemic Arithmetic and all three schemas below:

1. For all i, j and all $\mathcal{L}$-formulas φ,

$$\mathcal{M} \vDash K_i \ \mathrm{ucl}(K_j\varphi \to \varphi). \tag{Truth}$$

2. For all jRi and all φ with $\mathrm{FV}(\varphi) = \{x\}$,

$$\mathcal{M} \vDash \exists e\, K_i \forall x (K_j\varphi \leftrightarrow W(e, x)). \tag{Code}$$

3. For all i, j and all such unary φ,

$$\mathcal{M} \vDash K_i \exists e \, \forall x (K_j \varphi \leftrightarrow W(e, x)). \tag{Mechanical}$$

Here φ^s is the result of replacing each free variable by the canonical numeral for its value under s, and ucl is universal closure. The bound index variable is always chosen fresh. No transitivity, decidable downward closure, recursive rank, finite branching, bounded height, modal-depth bound, or restriction on arithmetic complexity is assumed.

The distinction between (Code) and (Mechanical) is the placement of the existential quantifier relative to K_i. The construction also gives the parameter version of (Mechanical) stated in Section 18.1.

2. Logical and computability preliminaries

2.1. The base logic

Alexander's modal structures satisfy free-variable independence, invariance under alphabetic changes of bound variables, and capture-free variable-for-variable substitution. They do not in general satisfy arbitrary term substitution through a modal operator. We use his sound and complete base consequence relation, written $\vdash$ below. See [A, §1.2, especially Definition 1.2.3 and Theorem 1.2.6].

For closed premises, the finite-certificate form of completeness is:

$$\Sigma \vdash \theta \quad \Longleftrightarrow \quad \text{some } \sigma_1, \ldots, \sigma_m \in \Sigma \text{ make } \sigma_1 \to \cdots \to \sigma_m \to \theta \text{ base-logically valid.} \tag{L1}$$

The empty list is allowed. Validity is r.e. in any effective countable modal language. One can enumerate its certificates using the ordinary first-order encoding in which each modal formula is a predicate on its ordered free-variable tuple, with the alphabetic and weak-substitution axioms BL1 and BL2.

All raw theorem predicates below reject nonclosed formulas. A consequence-closed theory of closed sentences is invariant under alphabetic changes, since such changes give base-logically equivalent sentences.

Write $\mathrm{PA}^{\mathcal{L}}$ for ordinary PA with induction extended to every formula of the displayed modal language. Its induction instances are

$$\mathrm{ucl}\left[\varphi(x \mid 0) \to (\forall x(\varphi \to \varphi(x \mid S(x))) \to \forall x \, \varphi)\right]. \tag{L2}$$

The remaining PA axioms are the usual arithmetic axioms. This is Alexander's Definition 1.2.19.

2.2. Assigned validity and numerical instantiation

For every base-logical validity η, include each closed numerical instance η^s as an **Assigned Validity** axiom. This is a substantive axiom scheme: a numerical instance of a base-logical validity need not itself be valid in every base-logical structure [A, Example 1.2.13].

All standard-arithmetic structures constructed below have the numerical-assignment property

$$M \vDash \eta[s] \quad \Longleftrightarrow \quad M \vDash \eta^s. \tag{L3}$$

Consequently they satisfy Assigned Validity.

A useful permitted numerical-instantiation argument is the following. Choose a fresh variable y. The variable-instantiation implication

$$\forall x\, \eta \;\rightarrow\; \eta(x \mid y)$$

is base-logically valid. Numerically close this implication, assigning y the desired number and closing any remaining free parameters. Assigned Validity supplies the resulting implication as an axiom. For several universal quantifiers, iterate this procedure. This derives the concrete instances used below without assuming a general modal term-substitution rule.

2.3. Arithmetic enumeration

Fix an effective syntax and a standard acceptable enumeration of programs. Choose $W(e, x)$ by a finite-computation representation, so that in standard arithmetic

$$\mathbb{N} \vDash W(\bar{e}, \bar{n}) \quad \Longleftrightarrow \quad n \in W_e,$$

and every true concrete instance $W(\bar{e}, \bar{n})$ is provable in PA. The latter follows by exhibiting the finite accepting computation and verifying its arithmetic trace. Only this positive trace property is used; PA is not required to decide nonmembership.

An r.e. sentence theory has an effective unary-column compiler: given a formula φ with sole free variable x, a program enumerates the numbers n for which the theory proves $\varphi(x \mid \bar{n})$. Dovetailing finite consequence certificates gives this program. The compiler is uniform in a program enumerating the theory and the formula.

These elementary computability facts, together with the *s-m-n* theorem and Kleene's recursion theorem, are the computability inputs to the source construction.

2.4. Knowing machines and self-doubting Epistemic Arithmetic

For each operator K_i, the pre-closure knowledge axioms are universal closures of knowledge of actual base-logical validities, distribution of knowledge over implication, positive introspection, and truthfulness. Self-doubting knowledge adds knowledge of each validity, distribution, and introspection axiom. Self-doubting Epistemic Arithmetic further includes every modal PA axiom and knowledge of that axiom [A, Definitions 1.2.9 and 1.2.19]. The word *self-doubting* describes the background axiom set: knowledge of one's own truthfulness is an additional requirement, rather than a background axiom. The (Truth) schema in Section 1 supplies that additional requirement for the family constructed here.

A family of theories is a family of knowing entities when its standard consequence interpretation satisfies these knowledge axioms; it is a family of knowing machines when each theory is r.e. [A, Definitions 1.2.15 and 1.2.17]. Our conclusion includes joint enumerability of the entire family and truth of every accepted sentence in its actual interpretation. Throughout the proof an **owner** is the agent whose modal operator is being interpreted, and a **holder** is the agent whose theory contains a displayed axiom. They may be different agents.

3. The fixed source theories

We now specify the theories whose soundness must be proved. The **original agents** are the given agents indexed by natural numbers. The additional owner $*$ is an auxiliary agent used to supply a common theory. The common theory is fixed independently of the hierarchy; each original theory receives codes for its columns and for the columns of its direct subordinates. Every program numeral selected in this construction remains fixed throughout the semantic argument.

Temporarily enlarge the language by one operator K_*. Let

$$I = \{*\} \sqcup \mathbb{N}, \qquad \mathcal{L}_* = \mathcal{L} \cup \{K_*\}.$$

All formula schemes in this section range over the complete enlarged language, including formulas that contain K_*.

3.1. Background and own-prefix closure

For each holder $a \in I$, let B_a consist of the following closed sentences:

- All axioms of $\mathrm{PA}^{\mathcal{L}_*}$.
- All Assigned Validity sentences.
- Own validity $\mathrm{ucl}(K_a\varphi)$, for every actual base-logical validity φ.
- Own deduction
$$\mathrm{ucl}(K_a(\varphi \to \psi) \to (K_a\varphi \to K_a\psi)).$$
- Own positive introspection
$$\mathrm{ucl}(K_a\varphi \to K_aK_a\varphi).$$
- All-agent reflection
$$\mathrm{ucl}(K_b\varphi \to \varphi), \qquad b \in I.$$

For a set X of closed generators, define

$$\mathrm{cl}_a(X) = \mathrm{Cn}_{\mathrm{BL}}\{K_a^n\sigma : \sigma \in X,\ n \in \mathbb{N}\}, \tag{S1}$$

where $K_a^0\sigma = \sigma$, and $\mathrm{Cn}_{\mathrm{BL}}$ here retains closed consequences.

The following syntactic fact will be used repeatedly.

Own sentence necessity. If $S = \mathrm{cl}_a(X)$, with $B_a \subseteq X$, and $S \vdash \sigma$, then $S \vdash K_a\sigma$.

Proof. Take a finite certificate with prefixed-generator premises $\eta_1, \dots, \eta_m$ and valid closed implication $V = \eta_1 \to \cdots \to \eta_m \to \sigma$. The next prefix puts $K_a\eta_j$ among the axioms. Own validity gives K_aV, and repeated own deduction gives $K_a\sigma$. This uses no soundness assumption. □

3.2. The common auxiliary theory

We call the common theory an auxiliary **core** because all original theories will inherit it. This is a theory of modal arithmetic; the ordinal core introduced in Section 6 is the separate structure used to label proof stages.

For a formula φ, a chosen input variable x, and a fresh variable $e \neq x$, put

$$\mathrm{MC}_b(\varphi; x) = \mathrm{ucl}\left[\exists e\, \forall x(K_b\varphi \leftrightarrow W(e, x))\right]. \tag{S2}$$

Thus the other free parameters of φ are universally quantified outside $\exists e$. The formula need not have x as its only free variable, or even have x free.

Define

$$U = \mathrm{cl}_* \left(B_* + \{ \mathrm{MC}_j(\varphi; x) : j \in \mathbb{N},\ \varphi,\ x \} \right). \tag{S3}$$

The primitive mechanicalness targets are the original agents j. In particular, this definition includes no primitive $\mathrm{MC}_*(\varphi; x)$ generators. Arguments φ may still contain K_*.

The theory U is r.e. and independent of R. Fix once and for all a total computable compiler d such that, for every unary φ,

$$n \in W_{d(\varphi)} \quad \iff \quad U \vdash \varphi(x \mid \bar{n}). \tag{S4}$$

This compiler is chosen before the hierarchy-dependent recursion fixed point.

3.3. The original theories and their fixed point

Let q be a program intended to enumerate pairs (i, σ) of original owners and their closed theorems. Uniformly from q, j, φ, obtain a program $c(q, j, \varphi)$ enumerating

$$\{ n : q \text{ enumerates } (j, \varphi(x \mid \bar{n})) \}.$$

For a prospective q, set

$$\begin{aligned} T_i(q) = \mathrm{cl}_i(B_i &+ \{ C_{*,\varphi}(d(\varphi)) : \mathrm{FV}(\varphi) = \{x\} \} \\ &+ \{ C_{j,\varphi}(c(q, j, \varphi)) : jRi,\ \mathrm{FV}(\varphi) = \{x\} \}), \end{aligned} \tag{S5}$$

where

$$C_{b,\varphi}(p) = \forall x (K_b \varphi \leftrightarrow W(\bar{p}, x)). \tag{S6}$$

The enumeration of R supplies positive edge witnesses. All other admissions, own prefixes, and finite logical consequences are effectively enumerated. Thus there is a total computable operation taking q to a program enumerating all pairs (i, σ) with $T_i(q) \vdash \sigma$.

Kleene's recursion theorem gives a q whose enumerated pairs are exactly those produced by this operation. Fix such a q, and set $T_i = T_i(q)$. Then

$$n \in W_{c(q,j,\varphi)} \quad \iff \quad T_j \vdash \varphi(x \mid \bar{n}). \tag{S7}$$

The source family $S_* = U,\ S_i = T_i$ is uniformly r.e. and consequence-closed.

This completes the source construction before any semantic argument. In every auxiliary model below, these same theories, source formulas, and program numerals remain fixed. Masking a semantic predicate does not rerun the fixed point or change a displayed W-index.

For convenient notation write

$$p(*, \varphi) = d(\varphi), \qquad p(j, \varphi) = c(q, j, \varphi) \quad (j \in \mathbb{N}). \tag{S8}$$

For each source code generator, its $p(w, \varphi)$ enumerates exactly the corresponding full source-theorem column.

4. Filtered models, shape agreement, and genericity

To prove source soundness without assuming it, we first interpret knowledge using a truth filter. We then prove that retaining source theories on a lower ideal preserves their soundness under all sound changes to the remaining predicates. The term **genericity** below names this precise preservation property.

4.1. Filtered truth

Write $\mathrm{Th}(S)$ for the predicate on closed sentences that holds exactly when S proves the sentence in the base logic.

A raw predicate family $P = (P_a)_{a\in I}$ assigns an alphabetically invariant predicate to closed $\mathcal{L}_*$-formulas, and rejects nonclosed formulas. Define a standard-arithmetic filtered structure $F(P)$ recursively by

$$F(P) \vDash K_a\varphi[s] \quad \iff \quad P_a(\varphi^s) \ \wedge \ F(P) \vDash \varphi[s]. \tag{F1}$$

This is recursion on formula syntax, so it requires no fixed-point or soundness assumption. Formula induction gives the base-logic structure conditions and the numerical-assignment property (L3).

Say that P_a is **raw-sound in** $F(P)$ if

$$P_a(\sigma) \quad \Longrightarrow \quad F(P) \vDash \sigma$$

for every closed σ. When every P_a is raw-sound, the truth conjunct in (F1) is redundant. Formula induction then identifies $F(P)$ with the unfiltered interpretation using exactly the predicates P_a. Our task is to prove this when $P_a = \mathrm{Th}(S_a)$.

Reflection is always true in the filtered model, before raw soundness has been proved.

4.2. Column admissibility and finite query restrictions

Call P **column-admissible** if it is alphabetic as above and, for every owner a, formula φ, variable x, and numerical parameter assignment s, there is an e such that

$$n \in W_e \quad \iff \quad P_a(\varphi^{s[x:=n]}) \qquad (n \in \mathbb{N}). \tag{F2}$$

The actual source family has this property by uniform r.e. theoremhood. Temporary families need only satisfy the displayed per-column condition. A uniform computable selection of all its witnesses is not part of the definition.

The **shape** of a formula keeps logical constructors, modal owners, and arithmetic predicate symbols, and forgets terms and binder-variable names. Let

$$Q(\theta) = \{(a, \mathrm{sh}(\varphi)) : K_a\varphi \text{ occurs anywhere in } \theta\}. \tag{F3}$$

Occurrences below other modal operators are included. This is a finite hereditary set: retaining an argument shape includes every modal query occurring inside that shape. Numerical closure, alphabetic change, and term variants preserve shape and the query set.

Two raw families that agree on every closed formula of every owner/shape class in $Q(\theta)$ give the same truth value to θ, at every assignment. This follows by formula induction. The universal quantifier case needs agreement on all numerical instances of an argument shape, which is why agreement is not restricted to a finite list of literal subformulas.

A selection depending only on owner and shape preserves column admissibility. For a fixed numerical column, owner and shape do not vary with n; the selected column is therefore one fixed old r.e. column or the empty column. A piecewise choice between two column-admissible families has the same property. The selector itself need not be computable.

4.3. Whole-ideal genericity

Let R^+ be the transitive closure of R, and write

$$D_R(J) = \{i : i \in J \text{ or } iR^+j \text{ for some } j \in J\}.$$

For a finite R^+-antichain J, define $G(J)$ to be this statement:

> For every column-admissible P such that $P_* = \mathrm{Th}(U)$ and $P_i = \mathrm{Th}(T_i)$ for all $i \in D_R(J)$, if every original P_i outside $D_R(J)$ is raw-sound in $F(P)$, then the core and every predicate in the retained ideal are raw-sound in $F(P)$.

No effectiveness of $D_R(J)$ is required. The source predicates retained on it are the fixed full theorem predicates.

We will prove $G(\emptyset)$ directly and the other cases by well-founded induction. The stronger temporary-model class in (F2) is needed when a finite formula is reconstructed while an infinite lower ideal is retained literally.

5. Direct genericity of the auxiliary theory

Core genericity lemma. Suppose P is column-admissible, $P_* = \mathrm{Th}(U)$, and every original P_j is raw-sound in $F(P)$. Then P_* is raw-sound there. Thus $G(\emptyset)$ holds.

Proof. First verify the unprefixed core generators.

Arithmetic holds in the standard first-order part. Modal induction follows by ordinary induction on the set of numbers satisfying its formula in $F(P)$; the numerical-assignment property identifies the literal 0 and successor instances [A, Lemma 1.2.22 and Proposition 1.2.23]. Actual validities and Assigned Validity are true by the base-logic structure conditions and (L3).

For core own validity, if φ is valid then φ is true at every assignment, and U contains its numerical instance φ^s. These give both conjuncts of $K_*\varphi[s]$.

For core deduction, the raw theory U is closed under modus ponens. The truth conjuncts of its antecedents give the corresponding implication of truths. For core introspection, $U \vdash \varphi^s$ implies $U \vdash K_*\varphi^s$ by own sentence necessity. If the inner filtered modal formula is true, that same formula supplies the outer truth conjunct. All reflection generators hold directly by (F1). None of these verifications assumes core raw soundness.

Consider $\mathrm{MC}_j(\varphi; x)$, with j original. Fix values for the outer parameters. Column admissibility supplies an e enumerating

$$\{n : P_j(\varphi^{s[x:=n]})\}.$$

Original-owner raw soundness makes $F(P) \models K_j\varphi[s[x := n]]$ equivalent to this raw test. The chosen e therefore witnesses the code existential for every n. Freshness of e ensures its value does not alter the argument column. Taking every parameter assignment verifies the universally closed generator.

Arguments containing K_* cause no extra assumption: the stipulated original raw soundness already concerns their complete truth in this same $F(P)$.

Finally, if a core generator σ is true, every $K_*^n\sigma$ is true by induction on n. Its body is true by the preceding step and is a literal core axiom, hence passes the raw U-test. All prefixed generators are therefore true. Base-logical soundness and finite certificates make every U-theorem true.

This proves the lemma without a staged core argument. □

6. The effective finite-guard ordinal input

Let $(\mathrm{Ord}; <, \leq_1, \leq_2)$ be the true **pure** pattern-of-resemblance structure of order 2. The relations are the corresponding elementary-substructure relations in the pure language; no arithmetic operations are added to this ordinal language.

Write $\upsilon_1 = 1^\infty$. The entire pure core is the initial segment $[0, \upsilon_1)$. We use its effective presentation and the following exact properties.

Ordinal lemma. There is a countably infinite recursive label domain L, with a recursive well-order $<$ and recursive relations $\leq_1, \leq_2$, isomorphic to this core, satisfying:

1. Both resemblance relations are reflexive and transitive, and
$$\leq_2 \subseteq \leq_1 \subseteq \leq .$$
2. **Connectedness:** if $p \leq_1 r$ and $p \leq q \leq r$, then $p \leq_1 q$.
3. **Finite-pattern reflection:** if $\alpha \leq_1 \beta$, every finite $F \subset [0, \beta)$ has a map
$$f : F \longrightarrow [0, \alpha)$$
fixing $F \cap [0, \alpha)$ and preserving and reflecting equality, $<$, $\leq_1$, and $\leq_2$ on F.
4. **Finite tallness:** for every natural k, there are
$$h_0 < \cdots < h_k$$
with $h_i \leq_2 h_j$ for every $i < j$. Such a list, beginning at $h_0 = \nu$, is called a guard $G_k(\nu)$. Thus a rank-k guard has $k + 1$ points.
5. **Finite room:** suppose $G_{k+1}(\lambda)$ is given, X is a finite set with $x < \lambda$ and $x \leq_1 \lambda$ for all $x \in X$, and $N \in \mathbb{N}$. Then there are N lists
$$Y_j = (y_{j,0}, \dots, y_{j,k}), \qquad j < N,$$
each a rank-k guard, such that every point is below and $\leq_1$-below λ, every point lies above every member of X, and
$$\max Y_i < \min Y_j \qquad (i < j < N). \tag{O1}$$
The assertions for $X = \emptyset$ and $N = 0$ have their usual vacuous meanings.

Proof of the ordinal lemma. First work with the actual core. The inclusions, transitivity, and connectedness are the elementary-substructure and respecting-forest properties of the true pure relations [W21, pp. 5–6].

For finite reflection, use the points of $F \cap \alpha$ as parameters in the existential closure of the complete finite atomic diagram of F. Include the negations of false atomic facts and the inequalities of distinct points. This is a Σ_1 assertion in the pure language containing both resemblance relations, true below β. Reflection at $\alpha \leq_1 \beta$ gives its witnesses below α. The parameter entries are fixed, and the full diagram gives exactly the required finite isomorphism. The case $\alpha = \beta$ uses the identity. A finite map may be extended arbitrarily to other labels when a total syntactic relabeling is wanted.

Finite tallness follows from the isominimal realization of every realizable finite pattern in the core [W18, Corollaries 4.8 and 4.10; W21, p. 6]. For each k, a finite increasing pairwise-$\leq_2$ pattern of length $k+1$ is realizable; for example it occurs on the corresponding finite segment $\upsilon_2, \ldots, \upsilon_{k+2}$ of the global sequence in [W21, Theorem 1.8]. Its isominimal finite copy lies in the core. This produces each finite guard separately.

For room, write the given guard as

$$\lambda = h_0 < h_1 < \cdots < h_{k+1}.$$

Apply [W21, Lemma 1.7(2)] with $\alpha = h_0$, $\beta = h_{k+1}$, and

$$Y = \{h_0, \ldots, h_k\} \subset [\alpha, \beta).$$

Every $y \in Y$ satisfies $y <_1 \beta$. The lemma supplies a copy of Y below α, above the finite set X, preserving its pattern over X; all copied points are $<_1 \alpha$. This gives the first required rank-k copy. Enlarge X by that entire copy and apply the same lemma again. Repeating finitely many times gives complete-copy separation (O1). Every enlarged parameter set remains finite and consists of points $<_1 \alpha$. This includes $k = 0$; the set being copied is then a singleton. The last guard point h_{k+1} is deliberately excluded from Y.

The effective presentation is supplied by the elementary-recursive characterization of the core [W17, §5.1 and Corollary 5.8], using the corrected semantic characterization in [W21, Theorem 4.2 and Corollary 4.6]. The latter repairs an earlier semantic proof, so the corrected identification is part of the cited input. Order and both resemblance relations are recursive on the canonical notation domain. If that domain is an infinite recursive set of numerical codes, enumerate its members in ordinary numerical order and pull back the relations along this computable bijection with $\mathbb{N}$. This gives the required recursive presentation on a domain coded by all natural numbers. Every property above transfers along the isomorphism. □

Only finite witnesses from this presentation enter stage enumeration. Their existence can be used classically in the proof, and a needed finite witness can also be found by search through its decidable finite conditions.

A useful consequence of connectedness is that any set C of points each $\leq_1$-below the same top λ is coherent in its increasing order:

$$u, v \in C, \quad u \leq v \quad \Longrightarrow \quad u \leq_1 v. \tag{O2}$$

In particular this holds for all points of the room copies together with X.

Within the core, finite guards of unbounded rank cannot all lie in one fixed proper initial segment [W17, introduction; W21, p. 6]. Nothing in the proof chooses an infinite $\leq_2$-chain in that core.

7. Labels, recorded syntax, and recording maps

This section introduces the syntax of a temporary labelled proof stage. The active owners are the finitely many source agents whose soundness is currently being established. External owners are finitely many owners outside the retained lower ideal; their old predicates are already sound. All remaining owners are passive. Active operators receive ordinal labels. An external query can remain **live**, with its old predicate, or become a **ghost**, whose knowledge predicate is empty. The choice depends on the argument's syntactic shape. Recording adds these labels and choices; erasure removes them and returns to the source language.

Let L be the effective well-ordered label set, with relations $\leq_1, \leq_2$. We use reflexivity of $\leq_1$, the implication $\alpha \leq_1 \beta \Rightarrow \alpha \leq \beta$, connectedness of $\leq_1$, finite-pattern reflection along $\leq_1$, and the stated finite-room property. Write $G_r(\nu)$ for a list of $r+1$ points beginning at ν, strictly increasing and pairwise $\leq_2$. No infinite chain is assumed.

Fix finite sets $A, E \subseteq \mathbb{N}$, disjoint from one another, with the following extension side conditions:

- The core $*$ is neither active nor external.
- If $a \in A$ and wRa, then $w \notin A \cup E$.

In the genericity application A is the finite frontier J, and E lies outside its downward ideal, so these conditions follow from antichainness.

Recorded modal heads are K_d, K_b^ν, and Z_e, with erasures K_d, K_b, K_e, respectively. All arithmetic terms and all numerical constants are unchanged by erasure ϵ. A recorded formula is **allowed** when every indexed owner belongs to A, every unindexed owner lies outside A, and every ghost owner belongs to E. It is **globally admissible** when it is allowed and, at every indexed occurrence $K_b^\nu \eta$, every indexed label anywhere inside η is strictly below ν. Inspection descends through unindexed heads, ghosts, Boolean connectives, and quantifiers.

The finite support $\mathrm{supp}(\eta)$ is the set of all indexed labels in the whole formula. In particular, labels below live or ghost heads remain in its support. The notation $\mathrm{supp}(\eta) < \nu$ means that every point of this finite set is strictly below ν, and is vacuous when the support is empty. Numerical closure, partial numerical substitution, and alphabetic variation preserve support and global admissibility.

The shape $\mathrm{sh}(\phi)$ retains logical constructors, modal owners, and arithmetic relation symbols, but forgets terms and binder-variable names. Its rank $r(\phi)$ is its constructor count. The query set is

$$Q(\phi) = \{(b, \mathrm{sh}(\psi)) : K_b\psi \text{ is any modal occurrence in } \phi\}.$$

It includes occurrences below other modal heads. It is finite and hereditary: if $(b, s) \in Q(\phi)$, every query inside s lies in $Q(\phi)$.

Let H be any finite hereditary set of owner/shape queries. Let $\ell(b, s) \in L$ be a label table. Define $t = t_{H,\ell}$ recursively, leaving terms, binders and Boolean structure unchanged, and setting

$$t(K_b\psi) = \begin{cases} K_b^{\ell(b,\mathrm{sh}\,\psi)} t\psi, & b \in A, \\ K_b t\psi, & b \in E,\ (b, \mathrm{sh}\,\psi) \in H, \\ Z_b t\psi, & b \in E,\ (b, \mathrm{sh}\,\psi) \notin H, \\ K_b t\psi, & b \notin A \cup E. \end{cases}$$

The only relevant entries of ℓ for a finite formula or certificate form a finite table. It can be extended arbitrarily elsewhere. No globally admissible extension of the table is needed to establish validity preservation.

This construction has four immediate invariants:

1. $\epsilon(t\phi) = \phi$.
2. t preserves free variables and commutes with capture-free syntactic substitutions, alphabetic changes, and numerical closure. This includes the substitutions displayed in modal induction. It is a property of the translation, not a claim that these substitutions preserve base-logical validity.
3. All formulas of one source shape have the same recorded support, admissibility condition, and query-label assignments.
4. If an external occurrence is kept live, heredity makes its entire recorded argument ghost-free.

The proofs are structural inductions. At a modal occurrence, syntactic substitution or numerical closure does not change the argument shape, so it does not change the chosen head. The induction hypothesis identifies the transformed argument. The other constructors are unchanged.

Lemma 1: preservation of actual base-logical validity

Erasure, arbitrary label renaming, and every recording map $t_{H,\ell}$ preserve base-logical validity in their forward directions.

Proof. Pull an arbitrary target modal structure back along the chosen translation. Interpret a source modal application $K_b\psi$ by the target interpretation of its translated modal application, with translated argument. Because the translation preserves free variables, irrelevant-variable invariance is inherited. Because it preserves alphabetic equivalence and commutes with the permitted variable substitutions, the alphabetic and variable-substitution conditions on modal interpretations are inherited. The arithmetic structure is unchanged. Induction on formulas then identifies truth in the pulled-back structure with truth of the translation. A source validity is consequently true after translation in every target structure.

Equivalently, in Alexander's first-order encoding, a pure modal formula names a predicate on its ordered tuple of free variables. Each translation replaces that predicate name by the name of its translated modal formula without changing arity. Images of the alphabetic and variable-substitution axioms BL1/BL2 are instances of the same axioms. A finite first-order validity derivation can therefore be translated line by line. Predicate identification under erasure causes no problem.

This argument needs only source sharing by owner and argument shape. It does **not** require the recording map to make every formula in the entire infinite source language globally admissible. Global admissibility is checked separately on each formula admitted to the stage.

The converse is unavailable: validity of $\epsilon\eta$ says nothing by itself about validity of η. In particular, $K_e\psi \leftrightarrow Z_e\psi$ is not admitted on account of its tautological erasure.

Numerical-substitution distinction. The fact that a translation commutes with numerical closure does not assert that numerical closure preserves base-logical validity. Assigned-validity rows below supply numerical instances explicitly. None of the subsequent syntactic arguments uses unrestricted term substitution through modal operators.

8. The actual stage generator set

The stage theories are generated by explicit axioms, rather than defined by their intended semantic behavior. Background rows express the logical and arithmetic rules. Main rows translate an original code axiom while retaining its source numeral; arithmetic marker antecedents ensure that every guard point belongs to the row's literal support. Own prefixes then close each holder's axioms under labelled knowledge. The later soundness proof must verify exactly this generator set.

For each active holder $a \in A$, define a set Γ_a of closed recorded generators by the following rules. Take the recorded modal language to consist of precisely the allowed heads. Thus the cut predicates for active holders suffice to interpret every indexed head, while arbitrary formulas in this language need not be globally admissible.

Background rows

Admit each of the following whenever the entire displayed formula is globally admissible:

1. Every modal PA axiom in the recorded language, including induction with an arbitrary recorded formula.
2. η^s, for every actual base-logical validity η and every numerical assignment to its finitely many free variables.
3. $\mathrm{ucl}(K_a^\nu \eta)$, for every actual validity η.
4. Same-label deduction:
$$\mathrm{ucl}(K_a^\nu(\eta \to \zeta) \to (K_a^\nu \eta \to K_a^\nu \zeta)).$$
5. Introspection, for $\mu < \nu$:
$$\mathrm{ucl}(K_a^\mu \eta \to K_a^\nu K_a^\mu \eta).$$
6. Truthfulness $\mathrm{ucl}(O\eta \to \eta)$ for every allowed indexed, unindexed, or ghost head O.
7. Contextual collapse, for $\mu \leq_1 \nu$:
$$\mathrm{ucl}(C[K_b^\mu \eta] \leftrightarrow C[K_b^\nu \eta]), \tag{CC}$$
where $C[-]$ is any one-hole recorded formula context and both fillings are globally admissible.

There is no source-sharing requirement on these background rows. In particular, the two fillings of a contextual-collapse row may assign different labels to different occurrences of one original source shape. This is essential: contextual collapse changes one occurrence at a time.

Padded main-code rows

Let $a \in A$, and let $C_{w,\phi}$ be one of a's actual source code generators. By the side conditions, $w \notin A \cup E$, so its outer head remains K_w.

Choose one hereditary H and one source-shared label table ℓ for the whole source sentence and its distinguished atom $K_w\phi$. Require $tC_{w,\phi}$ to be globally admissible. Inspect the whole recorded distinguished atom $t(K_w\phi)$. At each live external occurrence $K_e\chi$ in it, impose the following on every indexed occurrence $K_b^\nu \beta$ inside χ:

- There is a displayed guard $G_{r(\epsilon\chi)}(\nu)$.
- Every label $\xi \in \mathrm{supp}(\beta)$ satisfies both $\xi < \nu$ and $\xi \leq_1 \nu$.

Inspection also descends under ghosts. A ghost occurrence itself imposes no external guard, but a live external occurrence below it does. The code head K_w is inspected and imposes no external guard because $w \notin E$.

Let B be a finite set containing every point of every displayed guard, including its base. Choose distinct arithmetic tautologies $\tau_1, \dots, \tau_m$, one for each listed point $h_1, \dots, h_m$ of B, with their shapes absent from the following finite forbidden pool: all subformula shapes of the source code sentence and all subshapes of the argument shapes in H. During a finite proof transformation the pool may be enlarged by the finitely many shapes of that certificate; choosing tags outside this larger pool also satisfies the admission rule. One explicit infinite supply is

$$\tau_0 = (\bot \to \bot), \qquad \tau_{n+1} = (\bot \to \tau_n).$$

Their ranks $2n + 3$ are distinct, so a finite forbidden pool is avoided by choosing sufficiently large indices. These indices are tags, not encodings of labels. When labels move, the tags do not move.

Admit the closed padded sentence

$$K_a^{h_1}\tau_1 \to (K_a^{h_2}\tau_2 \to \cdots (K_a^{h_m}\tau_m \to tC_{w,\phi}) \cdots). \tag{M}$$

The markers are holder-local. Their arguments are arithmetic and have empty label support, so padding preserves global admissibility. The exact support of (M) is

$$B \cup \mathrm{supp}(tC_{w,\phi}).$$

Thus every guard point is literal formula support. There are no mechanicalness main rows in these extension stages.

The source numeral $\overline{p(w,\phi)}$ in (M) is unchanged. It is not replaced by an index computed from $t\phi$, and it does not encode a fresh or renamed outside owner. The marker bundle may be empty when no guards are needed; in that case the padded row is just its main sentence.

Own-prefix closure

If $\sigma \in \Gamma_a$, $a \in A$, and $\mathrm{supp}(\sigma) < \nu$, admit $K_a^\nu \sigma$. Iterate this rule finitely. Every resulting generator is closed and globally admissible.

Set

$$\Gamma_a^{<\nu} = \{\sigma \in \Gamma_a : \mathrm{supp}(\sigma) < \nu\}.$$

The cut predicate on a closed formula η is $\Gamma_a^{<\nu} \vdash \eta$, where consequence is Alexander's actual base-logical consequence. It is false on nonclosed formulas. No admissibility condition is placed on arbitrary logical conclusions. In particular, this cut is not defined by intersecting source theorem-hood with an erasure condition.

Lemma 2: effectivity of the actual cut

The predicates $\sigma \in \Gamma_a$, $\sigma \in \Gamma_a^{<\nu}$, and $\Gamma_a^{<\nu} \vdash \eta$ are uniformly r.e. in their finite parameters.

Proof. Enumerate finite syntax, finite masks, finite tables, finite guard lists, finite marker bundles, and finite prefix trees. Allowed-head checks, heredity, shape equality, freshness and global support calculations are finite syntactic checks. Every order, $\leq_1$, and $\leq_2$ assertion has a positive enumerating certificate. Source-code admissions use the actual source schema and positive R-edge

certificates. Actual validity has an r.e. proof system. A cut consequence is witnessed by a finite list of admitted cut generators and a valid implication chain to its conclusion. All these certificates can be dovetailed. No negative reachability test, soundness test, or computable rank is used.

For any fixed recorded formula η, label ν, owner a, assignment s, and variable x, the numerical column

$$\{n : \Gamma_a^{<\nu} \vdash \eta^{s[x:=n]}\}$$

is therefore r.e. The finitely many values of s used by η are hard-coded, and $n \mapsto \eta^{s[x:=n]}$ is a computable syntax operation.

9. Erasure and the finite local prefix lemma

Lemma 3: source sentence necessity

For every source holder a, $S_a \vdash \chi$ implies $S_a \vdash K_a\chi$ for closed χ. This is the purely syntactic necessity lemma proved in Section 3.1; it applies to the fixed theories of Section 3 without a source-soundness assumption.

Lemma 4: erasure

If $\Gamma_a^{<\nu} \vdash \eta$ and η is closed, then $S_a \vdash \epsilon\eta$. The same assertion holds without a cut.

Proof. Each background row erases to a source theorem: modal PA and structural rows erase to their source instances; assigned validity erases to an assigned source validity by Lemma 1; reflection erases to source truthfulness. A contextual-collapse row erases to $\mathrm{ucl}(\zeta \leftrightarrow \zeta)$, an actual validity.

A padded main row erases to an implication chain with the original source code generator as conclusion. That generator is already a theorem of S_a, so propositional logic gives the padded implication, irrespective of its marker antecedents. A generated own prefix erases to an own prefix of a source theorem, and Lemma 3 applies.

Translate the finite valid implication of a cut proof by erasure using Lemma 1. Its erased premises are all source theorems by the preceding induction, so its erased conclusion follows. Numeric constants, source formulas and assignments have not changed.

Lemma 5: local prefixing of a derived cut consequence

Suppose $a \in A$, χ is closed and globally admissible, $\mathrm{supp}(\chi) < \nu < \delta$, and $\Gamma_a^{<\nu} \vdash \chi$. Then

$$\Gamma_a^{<\delta} \vdash K_a^{\nu}\chi.$$

Proof. Choose a finite certificate with admitted cut premises $\eta_1, \dots, \eta_m$. They are closed and globally admissible, and their whole supports lie below ν, including any markers or earlier prefix labels. The exact valid chain

$$V = \eta_1 \to \cdots \to \eta_m \to \chi$$

is consequently closed, globally admissible, and supported below ν. Its validity is the actual validity certificate, not a conclusion drawn from erasure. Own validity admits $K_a^{\nu}V$ below δ. Own-prefix closure admits each $K_a^{\nu}\eta_j$ below δ. Same-label deduction distributes K_a^{ν} through the chain, yielding $K_a^{\nu}\chi$. All new labels are ν, hence below δ.

This proof uses arbitrary globally admissible background rows. The premise list may contain different source-shared main-row tables, and its combined implication need not possess one common source table.

Lemma 6: markers can be discharged in their own cut

Every marker $K_a^h\tau$ with $a \in A$ is derivable in every cut strictly above h. Therefore a padded main row whose whole support lies below δ yields its unpadded main sentence in $\Gamma_a^{<\delta}$.

Proof. The marker itself is an own-validity row, because τ is an actual closed arithmetic validity with empty support. Repeated modus ponens discharges the marker antecedents of (M).

10. Syntactic contextual equivalence and cut collapse

Lemma 7: contextual substitution inside one cut

If $\mu \leq_1 \nu$, the two fillings

$$\eta = C[K_b^\mu \phi], \qquad \zeta = C[K_b^\nu \phi]$$

are globally admissible, and both supports lie below δ, then for every assignment s,

$$\Gamma_a^{<\delta} \vdash \eta^s \iff \Gamma_a^{<\delta} \vdash \zeta^s. \tag{1}$$

Proof. Numerically close the context, preserving the binders on the path to the hole. More precisely, start with the partial substitution assigning $s(x)$ to every free variable. On entering a quantifier binding x, remove x from that substitution; transform each fixed side formula using the current substitution. At the hole use the resulting substitution on ϕ. This produces a context C_s and a filler ϕ_s satisfying the literal identities

$$\eta^s = C_s[K_b^\mu \phi_s], \qquad \zeta^s = C_s[K_b^\nu \phi_s].$$

Both completed fillings are closed. They remain globally admissible and have the same supports as before. Apply the contextual-collapse admission rule to C_s, ϕ_s, μ, ν. Its universal closure is vacuous, so the admitted formula is exactly $\eta^s \leftrightarrow \zeta^s$, and it lies below δ. Propositional logic gives (1).

This proof deliberately does not infer numerical instances through modal syntax from an arbitrary universal closure. The closed equivalence is a new literal instance of the same contextual schema.

Lemma 8: finite-pattern preservation of generators

Let $F \subseteq L$ be finite and $f : L \to L$ preserve $<, \leq_1, \leq_2$ on F. If $\sigma \in \Gamma_a$ and $\mathrm{supp}(\sigma) \subseteq F$, then its label-renaming $f\sigma$ belongs to Γ_a.

Proof. Inspect the admission certificate.

- Allowed owners do not change. Every nesting inequality is preserved because its two endpoints occur in the formula support.
- Modal PA commutes with label renaming. Actual validity is preserved by Lemma 1, and assigned rows retain their assignment.
- Structural and truthfulness rows retain their syntax. Their label inequalities remain true.
- Both endpoints of a contextual-collapse row occur in its whole support, so its $\leq_1$ premise remains true. Rename the arbitrary context and filler.

- A main row retains its source code sentence, source numeral, owner, hereditary mask, tag formulas, and tag order. Replace its table by $f \circ \ell$. Every displayed guard point occurs among its markers, so it belongs to F. Renaming therefore preserves all guard-chain relations and all whole-argument $\leq_1$ conditions. Freshness refers to unchanged source shapes and unchanged tags.
- A prefix row remains a prefix row because its complete body and prefix label occur in the support, and every required strict bound is preserved.

Induction on the finite prefix certificate completes the proof. Forward relation preservation suffices for this argument.

Lemma 9: exact cut collapse

If $\alpha \leq_1 \beta$ and $\mathrm{supp}(\eta) < \alpha$, then

$$\Gamma_a^{<\alpha} \vdash \eta \iff \Gamma_a^{<\beta} \vdash \eta. \tag{2}$$

Proof. The forward implication is monotonicity of cut generators. For the reverse, take a finite certificate below β, and let F be the union of the supports of its conclusion and displayed premises. Every point of F is below β. In particular, all guard markers and earlier prefix labels are included.

Finite reflection at $\alpha \leq_1 \beta$ gives a pattern map f taking F below α and fixing its points already below α. Lemma 8 transports each displayed generator premise into the cut below α. Lemma 1 transports the valid implication under arbitrary label renaming. The conclusion is fixed because its support was below α, so the reflected certificate proves η.

The support F concerns the displayed premises and conclusion. The logical validity derivation can be renamed globally regardless of the labels in its intermediate lines; those intermediate lines need no ordinal-pattern conditions. Thus one does not incorrectly demand that arbitrary labels inside a validity derivation already lie below the original cut.

11. The filtered stage model and background truth before stage soundness

Let $P = (P_a)_{a \in I}$ be alphabetic raw predicates on closed source formulas. Its filtered source model $F(P)$ uses

$$F(P) \Vdash K_a \phi[s] \iff P_a(\phi^s) \ \wedge \ F(P) \Vdash \phi[s].$$

No soundness of P_a is assumed.

Define the stage model $M = M(P)$ by standard arithmetic and the following recursive modal clauses:

$$\begin{aligned} M \Vdash K_d \eta[s] &\iff P_d((\epsilon\eta)^s) \wedge M \Vdash \eta[s], \\ M \Vdash K_a^\nu \eta[s] &\iff \Gamma_a^{<\nu} \vdash \eta^s \wedge M \Vdash \eta[s], \\ M \Vdash Z_e \eta[s] &\iff \text{false}. \end{aligned} \tag{3}$$

These clauses apply to arbitrary recorded formulas, not merely source-shared lifts.

Lemma 10: the stage model is a legitimate base-logical structure

The raw stage predicates are alphabetic. The filtered clauses satisfy the modal structure conditions and the numerical-assignment law

$$M \vDash \eta[s] \iff M \vDash \eta^s[t]$$

for every s, t.

Proof. Erasure takes alphabetic variants to alphabetic variants, so live predicates inherit the property from P. Closed alphabetic variants are base-logically equivalent by the alphabetic axioms; consequence closure therefore makes each cut predicate alphabetic. Ghost predicates are empty.

Free-variable invariance, alphabetic invariance, and the legal variable-substitution law now follow by formula induction. At a modal head the raw test is applied to the numerically closed argument, and the relevant closed arguments are identical or alphabetic variants by the syntactic substitution laws; the truth conjunct follows from the induction hypothesis. Numerical closure obeys the same recursive identities and removes every free variable, giving the displayed assignment law.

This constructs M before any stage-cut soundness theorem. Thus actual validities are true in M.

Lemma 11: contextual collapse is true before stage soundness

For any admitted instance (CC), M satisfies its two fillings equivalently at every assignment.

Proof. Induct on the one-hole context C.

At the hole, global admissibility puts the support of ϕ below μ. Equation (2) equates the two raw proof tests $\Gamma_b^{<\mu} \vdash \phi^s$ and $\Gamma_b^{<\nu} \vdash \phi^s$. Their filtered truth conjunct is the same assertion $M \vDash \phi[s]$.

Boolean contexts preserve equivalence, and quantified contexts use the induction hypothesis at every updated assignment. At an unindexed outer head the two argument erasures are literally equal, so the raw P_d tests coincide; the inner truth conjuncts coincide by induction. At a ghost outer head both values are false.

At an indexed outer head $K_d^{\xi} D[-]$, both filled argument supports lie below ξ. Lemma 7, applied in holder d's cut below ξ, equates the two raw cut tests. The induction hypothesis equates their truth conjuncts. This use of Lemma 7 is syntactic membership of an already admitted contextual row, not its assumed truth. Hence the induction is noncircular.

Taking all assignments verifies the universal closure in (CC).

Lemma 12: every background row is true in M

Proof. Arithmetic equations hold in standard arithmetic. For induction, the truth set $\{n : M \vDash \eta[s[x := n]]\}$ satisfies ordinary induction on $\mathbb{N}$. Its base and successor instances are the literal schema instances because $S(\bar{n}) = \overline{n+1}$ and numerical closure commutes with the displayed 0- and successor-substitutions. This uses no unrestricted substitutivity of extensionally equal arbitrary terms inside modalities.

An assigned validity is true by actual validity in M and Lemma 10's assignment law. For own validity $K_a^{\nu}\eta$, the argument is valid and hence true. Its numerical closure η^s is an assigned-validity row below ν, since its whole support lies below ν; this supplies the raw proof test.

Same-label deduction follows from modus ponens in $\Gamma_a^{<\nu}$, together with the truth implications in the antecedent filters. For introspection, an acceptance of $K_a^\mu \eta$ gives both $\Gamma_a^{<\mu} \vdash \eta^s$ and truth of that modal atom. Lemma 5, with $\mu < \nu$, gives $\Gamma_a^{<\nu} \vdash K_a^\mu \eta^s$, the needed raw test for the outer head. Its truth conjunct is the accepted inner modal atom.

Every truthfulness row follows directly from the filter in (3). Contextual collapse follows from Lemma 11.

Each marker is also true before stage soundness: τ is true, and its empty-support validity is provable in $\Gamma_a^{<h}$, so $K_a^h \tau$ is true. Therefore a padded main sentence is true precisely when its unpadded code sentence is true.

12. Coherent changes of labels and fixed-label transport

Lemma 13: coherent lifts are interchangeable inside a cut

Let $t_1\theta, t_2\theta$ be two globally admissible lifts of the same source formula, with the same hereditary live/ghost mask. Assume their labels lie in a set C such that

$$u, v \in C,\ u \le v \quad \Longrightarrow \quad u \le_1 v,$$

and both supports lie below δ. Then, for every active holder a and assignment s,

$$\Gamma_a^{<\delta} \vdash (t_1\theta)^s \quad \Longleftrightarrow \quad \Gamma_a^{<\delta} \vdash (t_2\theta)^s.$$

Proof. Form the pointwise minimum table $\ell_0 = \min(\ell_1, \ell_2)$ on each source owner/shape. It is globally admissible: if both original child labels are below their corresponding parent labels, choose a coordinate realizing the minimum parent; the minimum child is at most the child in that coordinate and is therefore strictly below the minimum parent.

Lower $t_1\theta$ to the minimum lift occurrence by occurrence, children before parents. At each step the changed label moves from v to $u \le v$, with $u \le_1 v$ by coherence. Its children have already been lowered and lie below u. Every unchanged indexed ancestor is still above the old label v, and hence above u. All other labels are unchanged. Thus both intermediate contexts are globally admissible and stay below δ. Lemma 7 gives a cut equivalence for this one occurrence. Compose finitely many such equivalences.

Do the same independently for $t_2\theta$, then reverse its equivalence to the minimum lift. Temporary intermediate formulas can violate source sharing, but this does not violate any contextual-collapse admission condition.

Lemma 14: guarded fixed-label transport

Suppose $a \in A$, $G_{k+1}(\lambda)$, $r(\beta) \le k+1$, and a prescribed globally admissible lift $\beta' = t_{H,\ell}\beta$ keeps every external query of β live. Assume

$$\xi \in \mathrm{supp}(\beta') \quad \Longrightarrow \quad \xi < \lambda \text{ and } \xi \le_1 \lambda.$$

Then for every assignment s,

$$\Gamma_a^{<\lambda} \vdash (\beta')^s \iff S_a \vdash \beta^s. \tag{4}$$

Proof. Erasure gives the forward implication. For reverse transport one may replace H by $Q(\beta)$, since the two masks agree at every external query of β, and thus give the same target lift.

Choose a finite source certificate for β^s, with premises $K_a^{n_j}\sigma_j$, where each σ_j is a source background or code generator. Apply the mask $Q(\beta)$ uniformly to the entire certificate.

Define active depth D by counting only active modal heads, taking maxima at Boolean branches, and passing through quantifiers and nonactive heads. An active head over an argument of depth d will receive a canonical label c_d. Let

$$N = \sum_j D(K_a^{n_j}\sigma_j) + D(\beta) + 1.$$

Every premise and the target have depth below N.

Apply finite room at the guarded top λ, above the finite prescribed support. Obtain N separated copies $Y_0, \dots, Y_{N-1}$, each containing a rank-k guard, with every point below and $\leq_1$-below λ. Let $c_d = \min Y_d$. Separation gives $c_d < c_e$ for $d < e$; connectedness gives $c_d \leq_1 c_e$ for $d \leq e$. Every original goal point and every new copy point belong to the same $\leq_1$-coherent set below λ.

The canonical depth table is globally admissible on the whole finite certificate. An active descendant query inside an argument of depth d has strictly smaller depth and hence a smaller label. We now derive each canonical recorded premise.

Background generators. Modal PA is carried to a literal PA instance, because recording commutes with the substitutions in induction. An assigned validity is carried to the same assignment of an actually valid recorded formula by Lemma 1. Own validity is a listed recorded own-validity row. Truthfulness becomes truthfulness of the selected indexed, live, or ghost head. Introspection has two successive active-depth labels and is its listed recorded instance.

Deduction requires an additional contextual-collapse step. Write

$$\gamma = c_{D(\phi\to\psi)}, \quad \mu = c_{D(\phi)}, \quad \nu = c_{D(\psi)}.$$

The canonical image is

$$\mathrm{ucl}(K_a^{\gamma}(t\phi \to t\psi) \to (K_a^{\mu}t\phi \to K_a^{\nu}t\psi)).$$

Since $D(\phi \to \psi) = \max(D\phi, D\psi)$, we have $\mu \leq_1 \gamma$ and $\nu \leq_1 \gamma$. Begin with the admitted same-label deduction row at γ. Change the displayed occurrence over $t\phi$ from γ to μ, and the displayed occurrence over $t\psi$ from γ to ν, using two contextual-collapse equivalences. Both final and intermediate formulas are admissible: the relevant arguments were already below their own smaller depth labels. All introduced labels were already in the canonical mixed row, so no cutoff increases. The universal-closure binders can be included in the contexts; no numerical specialization is needed.

Code generators. Let $\sigma_j = C_{w,\phi}$. The target w is passive and nonexternal by the side conditions, so the canonical image has the correct literal main-code shape and unchanged numeral. A live external argument χ in this image survives only because its owner/shape occurs in $Q(\beta)$. Thus

$$r(\chi) < r(\beta) \leq k + 1, \qquad r(\chi) \leq k.$$

An active occurrence at depth d inside χ receives the first $r(\chi)+1$ points of Y_d as its guard. Every active label inside its own argument has smaller depth; connectedness supplies the required $\leq_1$ relation to c_d.

Let $D_j = D(\sigma_j)$. All guards needed by this main row can be taken inside $\bigcup_{d<D_j} Y_d$. Choose fresh marker tags for these finite points. If the source premise has no own prefix, every marker and every main label lies below λ, so Lemma 6 derives the canonical unpadded code below λ. If it has at least one own prefix, separation puts every point of $\bigcup_{d<D_j} Y_d$ below c_{D_j}, the first own-prefix label; derive the base code already in that smaller cut.

Own prefixes. The equality

$$D(K_a^n\sigma) = D(\sigma) + n$$

holds because $a \in A$. Starting from the base proof below $c_{D(\sigma)}$, repeatedly use Lemma 5 at the successive labels

$$c_{D(\sigma)}, c_{D(\sigma)+1}, \cdots, c_{D(\sigma)+n-1}.$$

Every body, including the guards and marker proofs used to derive it, is in the preceding cut. The final prefix label is below λ. This derives the full recorded source premise in $\Gamma_a^{<\lambda}$.

The uniformly recorded finite source implication is actually valid by Lemma 1. Combining the recorded premise proofs derives the canonical recorded target below λ. Lemma 13 then identifies it with the prescribed target because all their labels lie in the common coherent set and both lifts are admissible.

In fact, room placed every canonical label above every prescribed goal label, so the final step can use direct child-before-parent lowering rather than pointwise minima. This is a simplification, not an extra assumption.

The number N, the source certificate, and the reflected or auxiliary proof labels may depend on s. The guarded top, target table and target shape do not. Formula variants of the same source shape have the same rank, support and admissibility conditions, so the argument applies separately to every such variant. No uniform numerical proof-depth bound has been used.

13. Exact recovery from one finite room

Lemma 15: recovery at room-copy heads

Let $Y_0, \dots, Y_{N-1}$ be room copies below a top λ with guard $G_{k+1}(\lambda)$, and let $h_i = \min Y_i$. Suppose a ghost-free prescribed lift $t\beta$ is admissible, $r(\beta) \le k+1$, and every target label is some h_d with $d < i < N$. Then

$$\Gamma_a^{<h_i} \vdash (t\beta)^s \iff S_a \vdash \beta^s$$

for every active holder a and assignment s.

Proof. Erasure is forward. Lemma 14 proves the target below λ. Room gives $h_i \le_1 \lambda$, while separation places the target support below h_i. Lemma 9 collapses the finite proof to h_i, fixing every target label.

For a given source formula θ, choose $k = r(\theta)$, $N = D(\theta) + 1$, and one such room using finite tallness and room. Record every active head over an argument of depth d at h_d, and keep every external query of θ live. Every active query $(a, \operatorname{sh}\psi) \in Q(\theta)$ has $D(\psi) < D(\theta)$, and all labels inside its recorded argument have smaller indices. Lemma 15 gives

$$\Gamma_a^{<h_{D(\psi)}} \vdash t\psi \iff S_a \vdash \psi \tag{5}$$

for every closed formula ψ of that query shape. On nonclosed formulas both predicates are false. Thus (5) covers every term variant and every numerical instance in the class, not only the literal occurrence in θ.

If the original active predicates are full source theoremhood, (5), literal live erasure, and formula induction give

$$M(P) \vDash t\theta[s] \iff F(P) \vDash \theta[s]. \tag{6}$$

Only the finitely many query shapes of this θ are required.

14. Reconstruction with full retained predicates

The stage language is a temporary instrument. To apply a lower-ideal genericity hypothesis, we need a source-language model in which the full lower theories are still present. The following reconstruction retains those predicates in full, uses stage cuts only for the active query classes of the target, and keeps the required external queries. Its soundness conclusion covers every accepted sentence of these predicates.

The following statement is the precise form needed in the lower-rank argument.

Fix a source formula θ, a hereditary mask H, and an admissible recorded lift $t\theta$. Let $D_0 \subseteq I \setminus (A \cup E)$ be the retained set. It can be infinite. Assume every nonactive, nonexternal owner appearing in $Q(\theta)$ either belongs to D_0 or has empty old predicate P_d.

Define a new original-language raw family Q' as follows, on closed formulas ψ:

$$Q'_d(\psi) = \begin{cases} P_d(\psi), & d \in D_0, \\ ((d, \operatorname{sh}\psi) \in Q(\theta)) \wedge (\Gamma_d^{<\ell(d,\operatorname{sh}\psi)} \vdash t\psi), & d \in A, \\ ((d, \operatorname{sh}\psi) \in Q(\theta) \cap H) \wedge P_d(\psi), & d \in E, \\ \text{false}, & \text{otherwise.} \end{cases} \tag{7}$$

All nonclosed formulas are rejected. Full retained predicates in the first branch are not restricted by shape and are not recovered from indexed cuts.

Lemma 16: reconstruction truth and column admissibility

For every formula ψ with $Q(\psi) \subseteq Q(\theta)$ and every assignment s,

$$F(Q') \vDash \psi[s] \iff M(P) \vDash t\psi[s]. \tag{8}$$

If P is column admissible, so is Q'.

Proof of truth. At an active modal occurrence the two raw tests agree by the second branch of (7) and commutation of recording with numerical closure. At a retained passive owner, erasure $t\chi \mapsto \chi$ makes the live raw test exactly $P_d(\chi^s)$. At a live external owner the third branch retains precisely this raw test; at a ghost it is false on both sides. At any remaining queried passive owner the hypothesis makes its old predicate empty, matching the final branch. The truth filters then agree by induction on the argument. Boolean connectives preserve equivalence, and quantifiers use the same induction hypothesis for every updated assignment.

Proof of admissibility. Alphabetic variation preserves shapes and commutes with recording; old predicates are alphabetic, and cut predicates are alphabetic by Lemma 10. Every branch in (7) is therefore alphabetic.

Fix one owner d, template χ, variable x, and parameter assignment s. All $\chi^{s[x:=n]}$ have the same shape, so the choice among branches of (7) is independent of n. The column is an old r.e. column, an empty column, or a column of one fixed recorded cut and one fixed finite recorded template $t\chi$. Lemma 2 supplies the last column. Hence every numerical parameter column is r.e. No jointly computable choice of ℓ, D_0, or program indices has been asserted or needed.

Lemma 17: full soundness of the new outside predicates

In addition to the preceding hypotheses, assume:

- **S:** For every $(a, q) \in Q(\theta)$ with $a \in A$, the raw predicate $\Gamma_a^{<\ell(a,q)} \vdash -$ is sound in $M(P)$ on every closed recorded formula, including nonadmissible conclusions.
- **E:** Every old P_e, $e \in E$, is fully sound in $F(P)$.
- **X:** For every $(e, \operatorname{sh}\chi) \in Q(\theta) \cap H$ with $e \in E$, every $(a, \operatorname{sh}\beta) \in Q(\chi)$ with $a \in A$, and every closed source formula ψ with $\operatorname{sh}\psi = \operatorname{sh}\beta$,

$$\Gamma_a^{<\ell(a,\operatorname{sh}\beta)} \vdash t\psi \quad \Longleftrightarrow \quad P_a(\psi).$$

  The query set $Q(\chi)$ depends only on its shape, so any representative χ of that shape gives the same condition.

Then every Q'_d with $d \notin D_0$ is fully sound in $F(Q')$.

Proof. A retained active acceptance $Q'_a(\psi^s)$ gives a cut proof of $(t\psi)^s$. The corresponding label occurs in $t\theta$ because the owner/shape class was retained. S gives $M(P) \vDash t\psi[s]$; (8) gives $F(Q') \vDash \psi[s]$. This argument applies to every accepted formula of that shape.

For an external acceptance, $(e, \operatorname{sh}\psi)$ lies in both $Q(\theta)$ and H, and $P_e(\psi^s)$ holds. E gives $F(P) \vDash \psi[s]$. Heredity makes every external head inside $t\psi$ live. At every nested active query, X identifies the new raw test with P_a; at retained passive heads the old predicates are literal; at nested external heads the third branch retains the old test; every other queried passive predicate is empty on both sides. Thus Q' and P agree on every query shape of ψ, including all variants and numerical instances. Formula induction transfers its old truth to $F(Q')$.

All other outside predicates are empty. This proves full outside soundness. No predicate in D_0 was assumed sound.

How literal main guards supply X

At a live external occurrence $K_e\chi$, the main-row guard rule gives $G_{r(\epsilon\chi)}(\nu)$ at every indexed descendant $K_b^\nu\beta$, and puts the entire support of β strictly and $\leq_1$-below ν. Its source argument shape has rank strictly below $r(\epsilon\chi)$. The retained external subtree is ghost-free by heredity. Apply Lemma 14 at ν, with $k+1 = r(\epsilon\chi)$, to each formula of the descendant's source shape. Erasure supplies the reverse direction. If $P_b = \operatorname{Th}(S_b)$ on active owners, this is exactly X.

The hypotheses are shape invariants, so the lemma holds for the full infinite class of term variants rather than only the inspected literal occurrence.

15. Local main truth and the two independent inductions

The induction on the given hierarchy supplies soundness for strictly lower source agents. Inside one step of that induction, a second induction on ordinal support proves truth of the stage generators. Main-row truth uses the already sound smaller cuts together with a strictly lower hierarchy frontier. This section establishes that inner step under the stated outer hypotheses; Section 16 then discharges the outer hypotheses for every frontier.

The following argument supplies the inner induction step used in the proof of $G(J)$ in Section 16. Its only outer induction hypotheses are $G(J')$ for strictly lower frontiers, and the direct core lemma of Section 5.

Fix a finite frontier $J = A$, let P be column-admissible and retain the full core and all source theories in the downward ideal of J, and suppose outside predicates are sound, with only the finite E possibly nonempty. Suppose genericity has already been proved for every strictly lower nonempty finite frontier, and the direct auxiliary-core genericity lemma is available.

Consider a main row for active holder a with target $w = *$ or wRa, source argument ϕ, and table t. Collect w when original, together with all original passive owners occurring in ϕ that lie strictly below J. Take the maximal finite frontier J' of this finite set, and let

$$D_0 = \{*\} \cup \downarrow J'.$$

Here $\downarrow J' = D_R(J')$ denotes the entire lower ideal, including the frontier itself. Every member of J' lies strictly below some member of J. If $J' \neq \emptyset$, its maximum well-founded R-rank is strictly below that of J. If the finite set is empty, $w = *$ and core genericity is the appropriate base case.

Every original passive full owner named in ϕ, and w itself, lies in D_0. No member of D_0 is active or external, by antichainness and the choice of E. An old outside owner outside E is empty. These facts verify exactly the passive-owner condition for (7).

Assume all cuts at labels in the main row's support are already sound. These are S. Old external soundness is E. Literal guards supply X by the preceding paragraph. Lemmas 16–17 construct a column-admissible Q', retain the entire D_0 predicates, and make every outside predicate fully sound. Lower-rank genericity, or core genericity when J' is empty, therefore makes S_w sound in $F(Q')$.

For each n with $S_w \vdash \phi(\bar{n})$, equation (8) now gives $M(P) \models t\phi[x := n]$. The outer live head K_w has raw predicate exactly S_w's theoremhood, and the arithmetic side still uses the actual source numeral $p(w, \phi)$. Consequently both sides of

$$\forall x(K_w t\phi \leftrightarrow W(\overline{p(w,\phi)}, x))$$

coincide for every n: forward uses its raw proof test and the source-column equation; reverse uses that equation and the just-proved recorded-body truth. The markers are independently true by Lemma 12. This proves the padded main row.

Lemma 18: support induction yields soundness of every stage cut

Every Γ_a-generator is true in $M(P)$, and every cut predicate is fully sound there.

Proof. Prove generator truth simultaneously for all active holders by well-founded induction on the maximum of nonempty finite supports, with an initial empty-support case.

At empty support, there are no indexed queries or prefix labels. Background rows are true by Lemma 12. A main row is covered by the local argument above, with S and X vacuous where no indexed labels occur. No prefix row has empty support.

For a label ν, assume every generator whose whole support is strictly below ν is true. Every cut at a label $\mu \leq \nu$ is then sound: every generator below μ has support strictly below ν, and base-logical soundness transfers their truth to all finite cut consequences.

Take a generator with maximum support ν. A background row is already true. A main row is true by the local reconstruction argument, since cuts at every label in its support are now sound. A prefix $K_a^\mu \sigma$ has $\mu = \nu$ because its body support lies strictly below μ. The induction hypothesis makes σ true; its literal generator membership below μ supplies the raw cut test. The prefix is therefore true.

This proves every finite-support generator. Base-logical soundness now makes every cut fully sound on all its closed consequences, including consequences that are not globally admissible. No stage proof was used to assume the original family's soundness. The outer original-rank induction was used only for the strictly lower J' selected in the main-row argument.

16. Whole-ideal genericity and the actual family

We now discharge the outer induction hypotheses used in Section 15.

16.1. The outer induction

Since R is well founded, let

$$\rho(i) = \sup_{jRi}(\rho(j) + 1).$$

Then jR^+i implies $\rho(j) < \rho(i)$. For a nonempty finite antichain J, put

$$r(J) = \max_{j \in J} \rho(j).$$

This is a metatheoretic ordinal rank. It is never queried by an enumerator.

Prove $G(J)$, as defined in Section 4.3, by induction on $r(J)$ for nonempty J. The empty frontier is the separately proved core lemma $G(\emptyset)$. In particular, no assertion about a maximum of the empty set is used.

Fix a nonempty J and assume $G(J')$ for every nonempty finite frontier of smaller rank.

16.2. Reduction to finitely many nonempty outside owners

It suffices to prove $G(J)$ for families whose nonempty original outside predicates have their owners in a finite set E.

Indeed, let P be an arbitrary family satisfying the hypotheses of $G(J)$, and let θ be a theorem of a retained source owner. Define $\widehat{P}$ by retaining the entire core and ideal unchanged and setting, for outside owners,

$$\widehat{P}_b(\psi) = P_b(\psi) \ \wedge \ ((b, \operatorname{sh} \psi) \in Q(\theta))$$

on closed ψ. This family is column-admissible by Section 4.2. Its nonempty outside owners belong to the finite set of outside names occurring in $Q(\theta)$.

Every retained outside acceptance has an argument shape from $Q(\theta)$. All queries inside an argument of that shape also lie in $Q(\theta)$, by heredity. Thus $F(P)$ and $F(\widehat{P})$ agree on that argument. Old outside raw soundness transfers to $\widehat{P}$, making every outside predicate raw-sound in $F(\widehat{P})$.

The finite-outside version of $G(J)$ makes θ true in $F(\widehat{P})$. Agreement on $Q(\theta)$ makes it true in $F(P)$. Applying this argument separately to every retained theorem, including core theorems, proves the full $G(J)$. This semantic reduction does not change the source family, its theories, or any program numeral.

We may therefore fix a finite $E \subseteq \mathbb{N} \setminus D_R(J)$ containing all nonempty original outside owners.

16.3. Stage soundness at the current frontier

Set $A = J$, and build the extension stages of Sections 7–15. Their side conditions hold: the core is passive, E is external to the ideal, and every direct subordinate of an active owner is passive and nonexternal.

The local argument of Section 15 selects, from a main source-code argument, a finite set of strictly lower owners, including its code owner when that owner is original. Every member of its maximal frontier J' lies strictly below some member of J. Therefore, whenever $J' \neq \emptyset$,

$$r(J') < r(J).$$

The available outer induction hypothesis is exactly the lower genericity required there. When $J' = \emptyset$, Section 5 is available instead. Lemma 18 now proves that every stage cut is raw-sound in the stage model $M(P)$.

The induction order is thus fixed. Syntactic transport, cut collapse, and background truth were proved first. Main-row truth uses sound smaller stage cuts and genericity of a strictly lower original frontier. Neither operation assumes soundness of an active original source theory.

16.4. Recovery of the entire original ideal

First take an active owner $a \in J$ and an arbitrary closed theorem θ of T_a. Choose the finite room for θ from Lemma 15, and let $t\theta$ be its fully live room lift.

Apply Lemma 14 at the room top λ. It gives

$$\Gamma_a^{<\lambda} \vdash t\theta.$$

Stage-cut soundness makes $t\theta$ true in $M(P)$. Room recovery at every active query shape in θ, together with literal erasure at passive heads, identifies that truth with $F(P) \vDash \theta$. Thus every active source predicate is raw-sound in $F(P)$.

Now let $i \in D_R(J) \setminus J$ and take an arbitrary closed T_i-theorem θ. Choose a fully live room lift $t\theta$ as before. Form the finite set consisting of i and every original owner named in θ that lies strictly below J. Let J' be its R^+-maximal frontier, and retain

$$D_0 = \{*\} \cup D_R(J').$$

This frontier is nonempty, contains i in its downward ideal, and has smaller rank than J.

Apply the reconstruction of Section 14. Every full passive original owner named in θ belongs to D_0; any other nonactive, nonexternal queried owner has empty old predicate. All used stage cuts

are now sound, giving S. Old outside soundness gives E. Room recovery gives X at all active query shapes of θ, and hence in particular below every retained external argument. Lemmas 16–17 give a column-admissible family Q' that keeps the full source predicates on D_0 and has fully sound predicates outside D_0.

By $G(J')$, the unchanged T_i-predicate is sound in $F(Q')$, so $F(Q') \vDash \theta$. Reconstruction transfers this to $M(P) \vDash t\theta$, and room agreement transfers it to $F(P) \vDash \theta$. This proves raw soundness for every passive owner in the retained ideal.

These arguments are performed separately for each theorem. The ideal may be infinite, and unrelated theorems need not use related rooms or label tables.

Every original raw predicate is now sound in $F(P)$: those inside the ideal by the preceding arguments, and those outside by hypothesis. Apply the direct core lemma at this point to obtain raw soundness of U. This proves the finite-outside version of $G(J)$, hence its full version by Section 16.2. The outer induction is complete.

16.5. Soundness of the actual source family

Let $P_a^{\mathrm{all}} = \mathrm{Th}(S_a)$ for all $a \in I$. It is column-admissible. To prove original raw soundness, fix a closed T_i-theorem θ.

Take the finite set consisting of i and every original modal owner occurring in θ, and let J be its R^+-maximal frontier. Retain the full source core and $D_R(J)$, and empty every other original raw predicate. Call the resulting family P^J.

Empty outside predicates are raw-sound without any assumption about P^{all}. Thus $G(J)$ gives $F(P^J) \vDash \theta$. Every query in θ has retained owner, so query agreement gives

$$F(P^{\mathrm{all}}) \vDash \theta.$$

This proves raw soundness of every original source theory in the actual filtered model. Apply core genericity once more to get raw soundness of U.

All truth filters can now be removed. We obtain a genuine standard-arithmetic structure $\mathcal{M}_*$ with

$$\mathcal{M}_* \vDash K_a\varphi[s] \quad \Longleftrightarrow \quad S_a \vdash \varphi^s, \qquad a \in I, \tag{A1}$$

in which every source theorem is true.

17. The required schemas and the effective reduct

17.1. Epistemic Arithmetic and known truthfulness

Every original theory contains its displayed background and all finite own prefixes of that background. The source-family soundness theorem consequently verifies modal PA, knowledge of its axioms, own validity, deduction, introspection, their known versions, and reflection. These include all axioms of self-doubting Epistemic Arithmetic.

For every $i \in \mathbb{N}$, $b \in I$, and unrestricted enlarged-language formula φ, the closed reflection sentence

$$\sigma = \mathrm{ucl}(K_b\varphi \to \varphi)$$

belongs to T_i. By (A1), $\mathcal{M}_* \vDash K_i\sigma$. This proves full mutual known truthfulness in the enlarged language.

17.2. Known particular subordinate codes

For jRi and unary φ, let $e = c(q, j, \varphi)$. The exact closed sentence

$$C_{j,\varphi}(e) = \forall x(K_j\varphi \leftrightarrow W(\bar{e}, x))$$

is a T_i-generator. Thus

$$\mathcal{M}_* \vDash K_i C_{j,\varphi}(e).$$

Assigning the fresh variable e this numerical value witnesses (Code). Under (A1), the raw modal test closes that variable to exactly the source numeral appearing in the generator. This is an application of the numerical-assignment semantics, with no modal term-generalization principle.

The same argument gives each original agent a known particular code for every unary core column.

17.3. Entire-core inheritance

Inheritance lemma. Every closed U-theorem is a theorem of every T_i.

Proof. Suppose $U \vdash \sigma$, with σ closed, and choose a variable x. Put

$$\psi(x) = \sigma \wedge x = x.$$

This has exactly the one free variable x, and $U \vdash \psi(x \mid \bar{0})$. Equation (S4) gives $0 \in W_{d(\psi)}$. A finite positive computation trace supplies

$$\mathrm{PA} \vdash W(\overline{d(\psi)}, \bar{0}),$$

hence the same theorem in T_i.

The core-code generator in T_i is

$$\forall x(K_*\psi \leftrightarrow W(\overline{d(\psi)}, x)).$$

Use the Assigned Validity numerical-instantiation argument of Section 2.2 to obtain its 0-instance. The positive arithmetic fact gives $T_i \vdash K_*\psi(x \mid \bar{0})$. The closed core-reflection instance then gives $T_i \vdash \psi(x \mid \bar{0})$, hence $T_i \vdash \sigma$. □

This is a proof-theoretic inheritance argument, using the literal source-code numeral. It does not replace a code by one for a recorded or masked argument.

17.4. Known mechanicalness

For any original j, the sentence $\mathrm{MC}_j(\varphi; x)$ belongs to U. Inheritance puts it in every T_i, so (A1) gives

$$\mathcal{M}_* \vDash K_i\, \mathrm{MC}_j(\varphi; x). \tag{A2}$$

When $\mathrm{FV}(\varphi) = \{x\}$, this is precisely (Mechanical). When there are other free parameters, (A2) gives the stronger parameter schema recorded in Section 18.1.

17.5. Removing the auxiliary operator

For each original i, define

$$T_i^0 = \{\sigma : \sigma \text{ is a closed } \mathcal{L}\text{-formula and } T_i \vdash_{\mathcal{L}_*} \sigma\}. \tag{A3}$$

This is uniformly r.e.: enumerate original source theorems and filter by the decidable condition of containing no K_*.

It is consequence-closed in the original base logic. Indeed, a finite original-language proof from members of T_i^0 has a validity certificate that remains valid in the enlarged language, since every enlarged-language structure has an original-language reduct. Its premises are T_i-theorems, so its conclusion is a T_i-theorem and therefore belongs to T_i^0.

For every closed original-language sentence σ,

$$T_i^0 \vdash_{\mathcal{L}} \sigma \quad \Longleftrightarrow \quad T_i \vdash_{\mathcal{L}_*} \sigma. \tag{A4}$$

The reverse direction holds by the definition of T_i^0; the forward direction was just proved.

Consequently, formula induction identifies the original-language reduct of $\mathcal{M}_*$ with the actual consequence interpretation of (T_i^0). All the required original-language formulas, their modal induction instances, and their code sentences survive the reduct. In particular, the original source indices still enumerate the exact original unary columns by (A4).

Renaming T_i^0 as T_i proves the theorem of Section 1. □

18. Further consequences and scope

The same source family yields parameter mechanicalness and entire-theory inheritance. We then examine a self-mechanical extension of the auxiliary theory, construct a distinguished common agent, and specify the quantifier order and the self-code boundary of the result.

18.1. Known mechanicalness with parameters

For every original i, j and every original-language formula $\varphi(x, \vec{z})$, where $\vec{z}$ lists its other free variables, the construction gives

$$\mathcal{M} \models K_i \left[\forall \vec{z}\ \exists e\ \forall x (K_j \varphi(x, \vec{z}) \leftrightarrow W(e, x))\right]. \tag{P1}$$

The universal parameter quantifiers are inside K_i and outside $\exists e$. The witness index may depend on the values of $\vec{z}$. This is a known universally quantified sentence, not merely a separate known sentence for each externally fixed numeral tuple.

The proof is direct: (S3) includes that closed core generator, Section 17.3 transfers the entire sentence, and the reduct preserves it.

This is an additional explicit schema beyond the unary formulation of the target. No claim of its logical independence from the exact unary target plus the surrounding weak-modal arithmetic is needed or made. Moving parameter quantifiers through knowledge would require its own argument.

18.2. Entire-theory inheritance and transitive subordinate codes

If jRi, the inheritance proof of Section 17.3 works with T_j in place of U, using i's particular j-codes. Hence, first in the enlarged language and then in the reduct,

$$\mathrm{Th}(T_j) \subseteq \mathrm{Th}(T_i). \tag{P2}$$

Iterating along a finite path gives the same inclusion whenever jR^+i.

A particular code sentence for a lower agent, proved at one step of a path, is therefore inherited by every agent above that step. Thus the family has known particular codes for all R^+-subordinates. The theorem does not assert that R^+ is exactly the relation of all pairs for which such codes happen to be known; that would require additional negative assertions about nonedges.

This inheritance is a consequence of the 3-1-3 requirements with the standard W-representation and PA's positive computation certificates. In a general solution, own validity and the actual consequence semantics give each theory every Assigned Validity instance: validity of η makes $K_i\eta$ true at every assignment, hence $T_i \vdash \eta^s$. Known truthfulness makes the selected known code sentence true; a lower theorem then gives a true concrete positive W-fact, and the same padding and reflection argument transfers the theorem. Alexander explicitly discusses the propagation to transitive closure in [A, §2.6, p. 75].

Own prefix closure gives all finite own iterations of known assertions. Inheritance also gives appropriate nesting along subordinate chains: if kR^+jR^+i, σ is closed, and $T_k \vdash \sigma$, then

$$\mathcal{M} \vDash K_iK_jK_k\sigma.$$

These observations do not establish arbitrary mixed-agent common knowledge across unrelated agents.

18.3. A strict self-mechanical extension of the auxiliary theory

The auxiliary theory's generator set omits its own mechanicalness. To measure the effect of adding those generators, write $U^- = U$, and define in the same enlarged language

$$U^+ = \mathrm{cl}_*\left(B_* + \{\mathrm{MC}_b(\varphi; x) : b \in I,\ \varphi,\ x\}\right). \tag{P3}$$

This theory adds the core's own mechanicalness instances to the generator table of Section 3.2.

The inclusion $U^- \subseteq U^+$ is immediate, but strictness requires proof.

Strictness proposition. In the enlarged base logic,

$$\mathrm{Th}(U^-) \subsetneq \mathrm{Th}(U^+).$$

Proof. Define an arithmetic translation E by

$$E(K_*\psi) = E(\psi), \qquad E(K_j\psi) = \bot \quad (j \in \mathbb{N}),$$

and let it commute with arithmetic symbols, Boolean operations, and quantifiers. Interpret formulas over standard $\mathbb{N}$ by

$$M_E \vDash \varphi[s] \quad \Longleftrightarrow \quad \mathbb{N} \vDash E(\varphi)[s].$$

The translation respects alphabetic changes and capture-free substitution, and its truth depends only on free variables. Thus this is an admissible base-logical structure with the numerical-assignment property.

All B_*-generators are true. Modal induction is induction on a subset of standard $\mathbb{N}$. Core validity, deduction, introspection, and core reflection become their transparent classical counterparts. Original-owner reflection has a false antecedent. Actual validity and Assigned Validity hold as in Section 2.

Every original knowledge column is empty, so a fixed empty-program index witnesses every original-owner mechanicalness generator, including its parameter version. A core prefix preserves truth because K_* is transparent in this model. Hence $M_E \models U^-$.

Let

$$\chi(x) = \neg W(x, x).$$

Its arithmetic truth set is the diagonal nonhalting set

$$D = \{n : n \notin W_n\}.$$

This set is not r.e.: if $D = W_d$, then $d \in W_d$ would be equivalent to $d \notin W_d$. In M_E, the column for $K_*\chi(x)$ is exactly D. Therefore

$$\mathrm{MC}_*(\chi; x) = \exists e\, \forall x(K_*\chi(x) \leftrightarrow W(e, x)) \tag{P4}$$

is false in M_E.

By base-logical soundness, U^- does not prove (P4). The theory U^+ contains it as a generator. This proves strict inclusion. □

The separating structure is a base-logical countermodel, not a claim that its transparent core is itself an r.e. knowing machine. Arbitrary admissible structures are the correct models for this nonprovability argument.

In the source model constructed in Section 16.5, since U^- is r.e., its own actual χ-knowledge column is r.e., so (P4) is externally true there. But U^- does not prove (P4), so the actual U^- core does not know this instance of its own mechanicalness. The sentence is a generator of U^+.

If the source construction is carried out with U^- or U^+ as its auxiliary theory, denote the resulting original-agent theories by T_i^- and T_i^+, respectively. The strictness result concerns the auxiliary theories; it does not prove $\mathrm{Th}(T_i^-) \subseteq \mathrm{Th}(T_i^+)$. Changing the core changes the literal code numerals and the recursion fixed point, so the original generator sets are not simply nested.

18.4. A distinguished self-mechanical common agent

Given R, add a new least agent $*$, with

$$* \, R^* \, i \quad \text{for every original } i,$$

retain all original edges, and put no edges below $*$. The relation R^* is r.e. and well founded. Code its countable owner set effectively by $\mathbb{N}$.

Applying the theorem to R^* produces a distinguished agent $*$ that knows everyone's truthfulness and mechanicalness, including its own, and whose particular unary codes are known by every original agent. Thus such an additional common agent exists for every hierarchy in the theorem.

A fixed, R-independent theory for such an agent can be obtained from the source construction. Apply it to R^*, using a second temporary auxiliary owner †. The desired $*$ is now an original owner relative to †, so $U_\dagger$ includes mechanicalness for $*$.

Choose $U_\dagger$ and its compiler $d_\dagger$ once, independently of R, as in Section 3.2. The least original agent has no subordinate-code generators. Its enlarged theory is therefore the fixed theory

$$V_* = \mathrm{cl}_*\left(B_* + \left\{\forall x(K_\dagger\varphi \leftrightarrow W(\overline{d_\dagger(\varphi)}, x)) : \mathrm{FV}(\varphi) = \{x\}\right\}\right). \tag{P5}$$

It inherits $U_\dagger$, including mechanicalness for $*$. All other original agents know its particular codes along their added edges. Removing $\dagger$ by Section 17.5 gives a fixed distinguished theory V' in the desired language.

In fact,

$$\mathrm{Th}(U^+) \subseteq \mathrm{Th}(V'). \tag{P5a}$$

The reduct V' contains every B_*-generator, inherits mechanicalness for every remaining owner from $U_\dagger$, and is closed under own sentence necessity. It therefore contains every finite $*$-prefix of every generator of U^+, and consequence closure gives (P5a).

The constructed standard knowing-machine model interprets K_* by V'-theoremhood and makes every V'-theorem true. It consequently satisfies every axiom of U^+. This proves ordinary consistency of U^+ and gives a standard knowing-machine model of its axiom set.

The distinguished theory V' need not equal U^+. Replacing its raw $*$-predicate by exact U^+-theoremhood would require a separate soundness and genericity argument for that predicate. Formulas can contain negative occurrences of knowledge, so changing the theory changes their interpretation; subtheory inclusion alone does not justify the replacement.

18.5. Quantifier order in the finite-room argument

The selected-room statement proved in Sections 12–13 has the following quantifier order:

$$\forall\theta\ \exists\mathcal{R}_\theta\ \forall\psi\ \forall s\ \forall\pi\ \exists\mathcal{D}_{\theta,\psi,s,\pi}. \tag{P6}$$

Here $\mathcal{R}_\theta$ is one finite target room and its label table; ψ ranges over the target and the formulas in its finitely many required query-shape classes, including all term variants; s is a numerical assignment; π is a finite source certificate for the required instance; and $\mathcal{D}$ is the additional finite proof room and reflection data. The target table is fixed before every such variant and instance. The additional proof room may depend on the certificate.

This is the recovery statement used in the soundness argument: one target room handles every required query-shape class, while additional proof rooms may vary with numerical assignments and certificates. No bound on all certificate depths is needed. The temporary predicates reconstructed in Section 14 have r.e. numerical columns; a joint computable selector for their column indices is unnecessary. The original source family itself remains uniformly r.e. by Section 3.

18.6. The boundary for known particular self-codes

The theorem gives each agent knowledge of everyone's truthfulness and mechanicalness, including its own. A stronger particular-code requirement would give every agent a known particular self-code for every unary knowledge column while retaining known self-truthfulness. Reinhardt's negative theorem excludes that full combination under the relevant Epistemic Arithmetic assumptions [A, §1.1, pp. 1–2, 5, and 7]. In Alexander's classification, these excluded strengthenings include Questions 3-2-3 and 3-3-3.

This concerns the entire self-code schema. An agent may know a particular code for an individual trivial column without contradiction.

References

[**A**] Samuel A. Alexander, *The Theory of Several Knowing Machines*, doctoral dissertation, The Ohio State University, 2013. Relevant locations: §1.1 and Definition 1.1.1 for the classification and negative boundary; §1.2, Definitions 1.2.3, 1.2.9, 1.2.15, 1.2.17, and 1.2.19, Theorem 1.2.6, and Example 1.2.13 for the logic and machine semantics; §2.6, p. 75, for inheritance along transitive closure. Official dissertation record.

[**W17**] Gunnar Wilken, *Tracking chains revisited*, 2017 version. §5.1 and Corollary 5.8 give the elementary-recursive characterization used for the core. Primary text.

[**W18**] Gunnar Wilken, *Pure patterns of order 2*, 2018 version. Corollaries 4.8 and 4.10 give the finite-pattern realization and core results used in the finite-tallness argument. Primary text.

[**W21**] Gunnar Wilken, *Pure Σ_2-Elementarity beyond the Core*, 2021 version. Printed pp. 5–6 include the pure relations, Lemma 1.7, the core identification, and Theorem 1.8. Theorem 4.2 and Corollary 4.6 supply the corrected semantic characterization. Preprint record.